Evaluating ML Classification Models for Diabetes Risk Screening: Comparative Study

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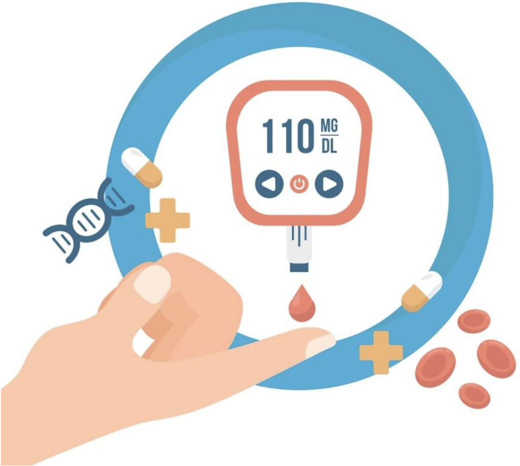
Diabetes is a growing global crisis

- **589M adults** with diabetes worldwide (2024 International Diabetes Federation)
- **~1 trillion** annual global spending in healthcare (International Diabetes Federation)
- **Type-2** diabetes accounts for majority, linked to lifestyle, environment & genetics

Can ML models reliably screen for diabetes risk?
Practical Screening Tool?



Our Approach

Dataset: https://www.kaggle.com/datasets/rabieelkharoua/diabetes-health-dataset-analysis

Full feature evaluation

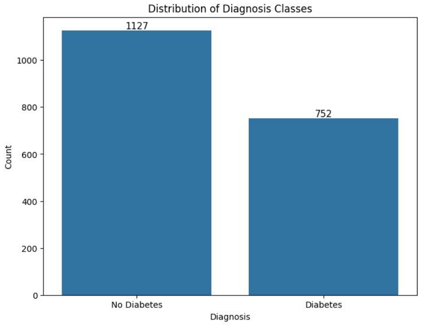
Accessible feature evaluation

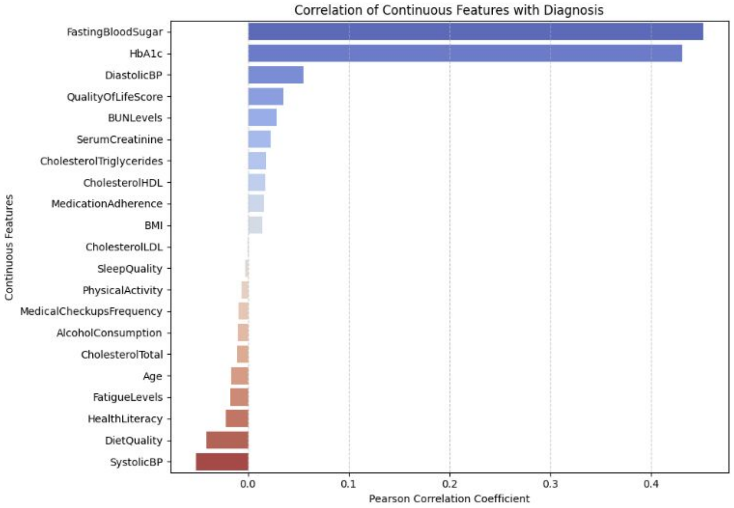
We train 4 models:

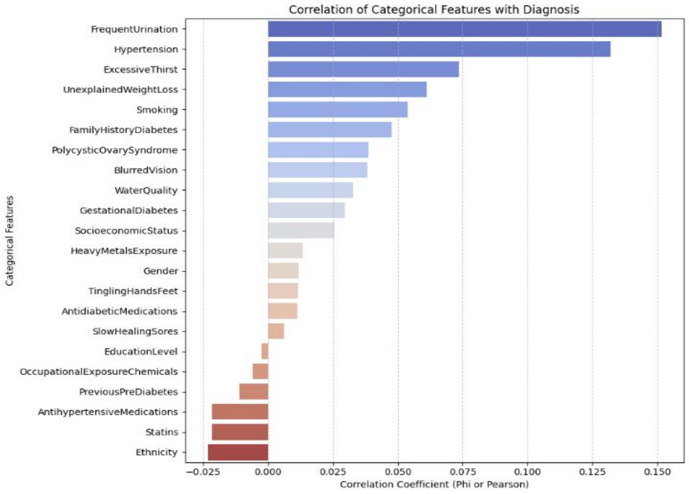
- ***Logistic Regression*** (class weighted - baseline)
- **Random Forest** (bagging, feature interactions)
- **XGBoost** (gradient boosting)
- **SVC** (non linear kernel)

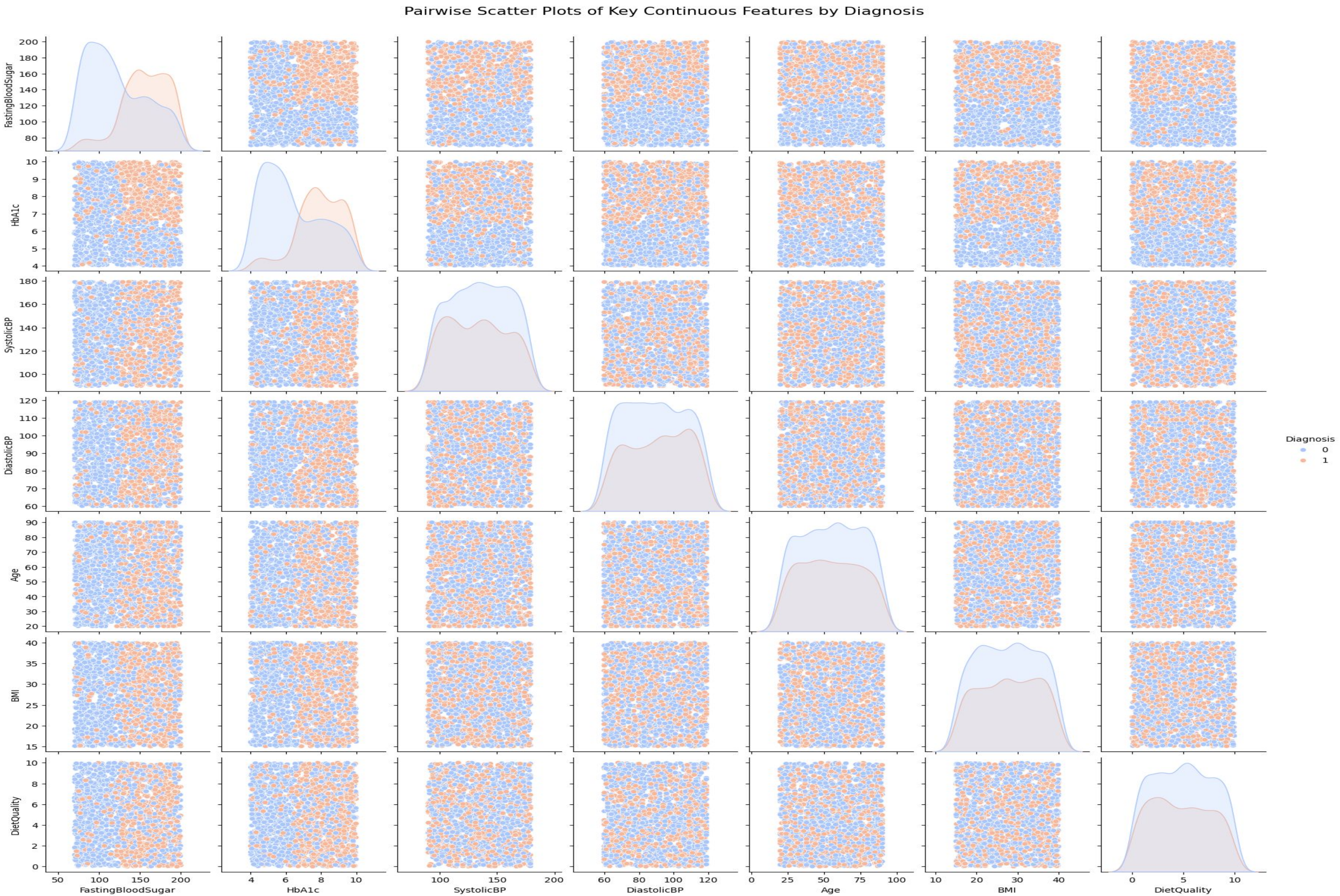
Exploratory Data Analysis

- 1,879 patient records
- 44 features: 23 categorical, 21 continuous, no missing values or duplicates
- Target: Diagnosis (Yes/No)








Pairwise Scatter Plots of Key Continuous Features by Diagnosis
FastingBloodSugar
HbA1c
SystolicBP
DiastolicBP
Age
BMI
DietQuality
Diagnosis
0
1

Data Preprocessing Pipeline

Before any model training:

1. Drop

Drop **PatientID** (identifier) and **DoctorInCharge** (constant).
Leaves **44** remain predictors for scaling.

2. Scale

StandardScaler from scikit-learn's library standardizes using zero mean, unit variance for all predictors.

3. Filter

Feature matrix is passed through a variance filter with a threshold of "0". This removes any constant

4. Test

ANOVA for continuous features, testing the means across the two diagnosis groups.
Chi-Square for categorical features ($\alpha = 0.05$), testing for independence with Diagnosis.

5. Split

75% training / **25%** test split + stratified **5-fold** cross-validation.
Fixed random seed.

Feature Significance

10 of 44 features pass α = 0.05

ANOVA		CHI-SQUARE	
FastingBloodSugar	F = 481.92 · <.0001	**FrequentUrination**	χ^2 = 42.36 · <.0001
HbA1c	F = 428.49 · <.0001	**Hypertension**	χ^2 = 31.94 · <.0001
DiastolicBP	F = 5.73 · 0.0168	**ExcessiveThirst**	χ^2 = 9.78 · 0.0018
SystolicBP	F = 5.05 · 0.0248	**UnexplainedWeightLoss**	χ^2 = 6.61 · 0.0102
		Smoking	χ^2 = 5.20 · 0.0225
		FamilyHistoryDiabetes	χ^2 = 4.05 · 0.0443

Tests consider each feature in isolation -> keep all 44 for model training!

Model Training & Evaluations

Models:

- Logistic Regression (43 multipliers, liblinear)
- Random Forest
- XGBoost (Gradient Boosting)
- SVC (RBF kernel)
  - sci-kit default is RBF kernel

Evaluation:

- Accuracy, Precision, Recall, F1-Score, Specificity, and ROC AUC
- 75/25 single-split
- Stratified 5-fold cross validation

Which one is the best? Linear or Non-linear?

Linear vs Non-Linear Models (Full Feature)

5-fold cross validation results

Model	Accuracy	Precision	Recall	F1-Score	ROC AUC
XGBoost	**93.4%**	**93.6%**	**89.6%**	**91.6%**	**0.958**
Random Forest	90.6%	94.4%	81.5%	87.4%	0.955
SVC	83.6%	78.2%	81.9%	80.0%	0.909
Logistic Regression	82.7%	76.3%	82.6%	79.3%	0.905

- **Non-linear beats Logistic Regression**
- **XGBoost is the strongest** overall
- Log Regression and SVC score lower but higher Recall than Random Forest

Full Feature vs Accessible Feature

Accessible features (self-reported)

- **Demographic:**: Age, Gender, Ethnicity, SocioeconomicStatus, EducationLevel, BMI, Smoking, AlcoholConsumption, PhysicalActivity, DietQuality, SleepQuality
- **Symptoms:** FrequentUrination, ExcessiveThirst, UnexplainedWeightLoss, BlurredVision, SlowHealingSores, TinglingHandsFeet, Hypertension*
- **Medical:**: FamilyHistoryDiabetes, GestationalDiabetes* PolycysticOvarySyndrome*, PreviousPreDiabetes
- **Environmental**: HeavyMetalsExposure, OccupationalExposureChemicals, WaterQuality

Model	Full Recall	Accessible Recall
XGBoost	89.6%	23.5%
Random Forest	81.5%	22.7%
SVC	81.9%	45.7%
Logistic Regression	82.6%	51.2%

Hyperparameter Tuning (Full Feature Set)

Grid Search performed on all models

Result: baseline models were already effective

Slight Performance Gains:
- XGBoost
- Logistic Regression

Negligible Performance Gain:
- SVC

Slight Performance Drop:
- Random Forest

Model	Best Parameters	Base ROC AUC	Tuned ROC AUC	Δ
XGBoost	**learning_rate: 0.2, max_depth: 7, n_estimators: 200, subsample: 1.0**	95.79%	95.98%	**+0.19%**
Random Forest	**max_depth: 10, min_samples_leaf: 2, min_samples_split: 5, n_estimators: 200**	95.54%	95.34%	**-0.20%**
Logistic Regression	**C: 0.1, penalty: 'l1', solver: 'liblinear'**	90.53%	90.95%	**+0.42%**
SVC	**C: 1, gamma: 'auto', kernel: 'rbf'**	90.94%	90.94%	**+0.0006%**

Hyperparameter Tuning (Accessible Feature Set)

Grid Search performed on all models

Result: best-performing model differs from baseline

Moderate Performance Gains:
- SVC
- Random Forest

Slight Performance Gain:
- XGBoost

Negligible Performance Drop:
- Logistic Regression

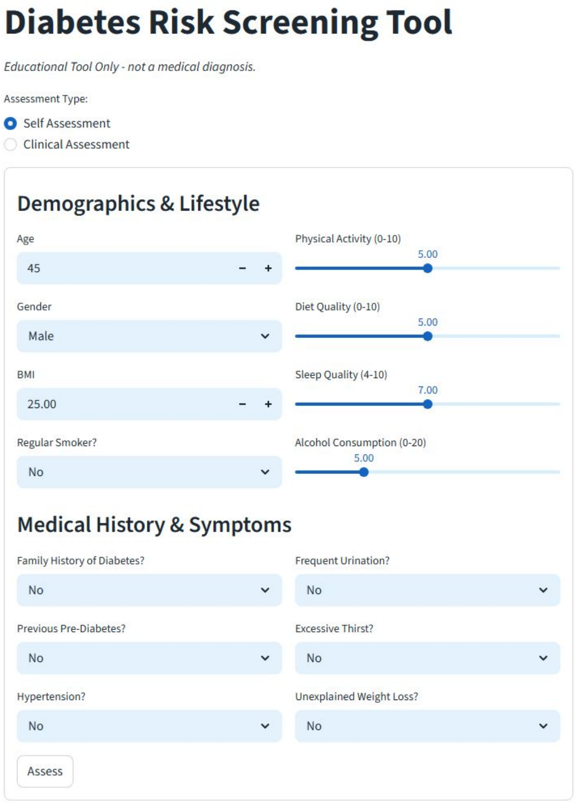
Model	Best Parameters	Base ROC AUC	Tuned ROC AUC	Δ
SVC	**C: 0.1, gamma: 'scale', kernel: 'rbf'**	59.12%	60.64%	**+01.52%**
Logistic Regression	**C: 0.1, penalty: 'l1', solver: 'liblinear'**	59.94%	59.89%	**-0.044%**
XGBoost	**learning_rate: 0.01, max_depth: 3, n_estimators: 100, subsample: 0.8**	58.04%	58.73%	**+0.69%**
Random Forest	**max_depth: 10, min_samples_leaf: 4, min_samples_split: 2, n_estimators: 100**	57.01%	58.41%	**+1.40%**

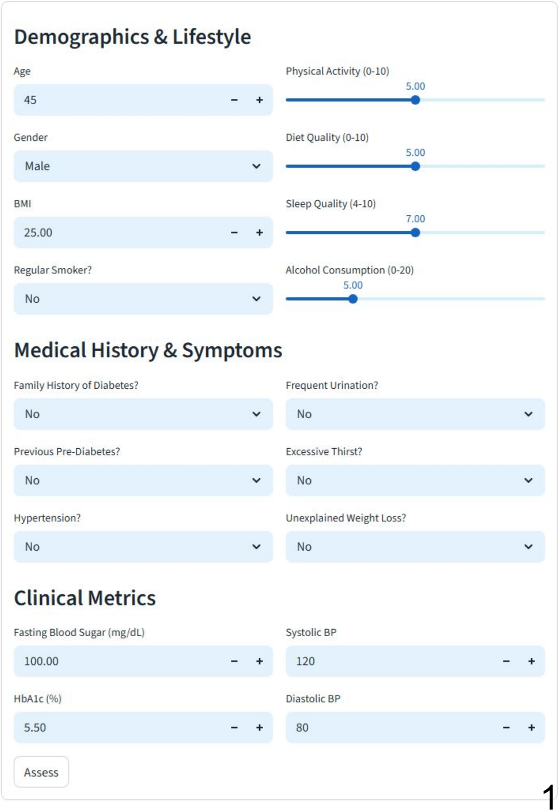
Web Application

Live Demo:
diabetes-screening.streamlit.app

Two screening options route to their corresponding models by feature set:

1) Self Assessment (Accessible)
2) Clinical Assessment (Full)





Conclusion

XGBoost is the best model for full clinical screening

F1 = 0.916 · ROC-AUC = 0.958. Non-linear models outperform the linear baseline by ~0.12 F1.

Accessible features alone are not enough

Best ROC-AUC drops to 0.606 and recall falls below 0.65, even after tuning.

Hyperparameter tuning is not the bottleneck

Gains are ≤ 0.004 ROC-AUC. Better features or more data will move the needle, not more grid search.

Proposed Solution: Deploy as a two-stage workflow

Self-report screen prompts users to obtain labs; clinical model delivers the high-accuracy risk estimate.

***ML can play a role** in early diabetes risk screening, but clinical data remains essential!*

Thank you.
Questions?

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